# Scatterplots and Correlation

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Statistics

# Introduction

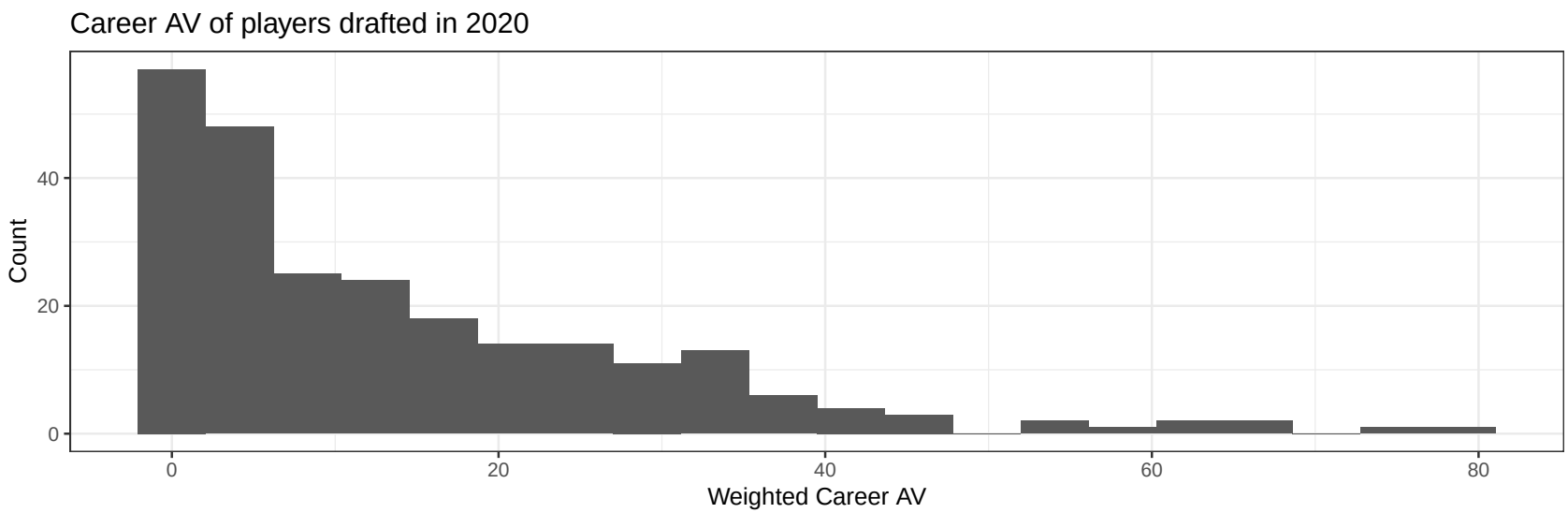
So far we've focused on *one-sample* and *two-sample* data:

1. One-sample data - is our sample consistent with a hypothesized population mean or proportion?
2. Two-sample data - does our sample show evidence of a difference between two populations (groups)?

These analyses have involved various combinations of *binary categorical* and *quantitative* variables. Which combination haven't we covered?

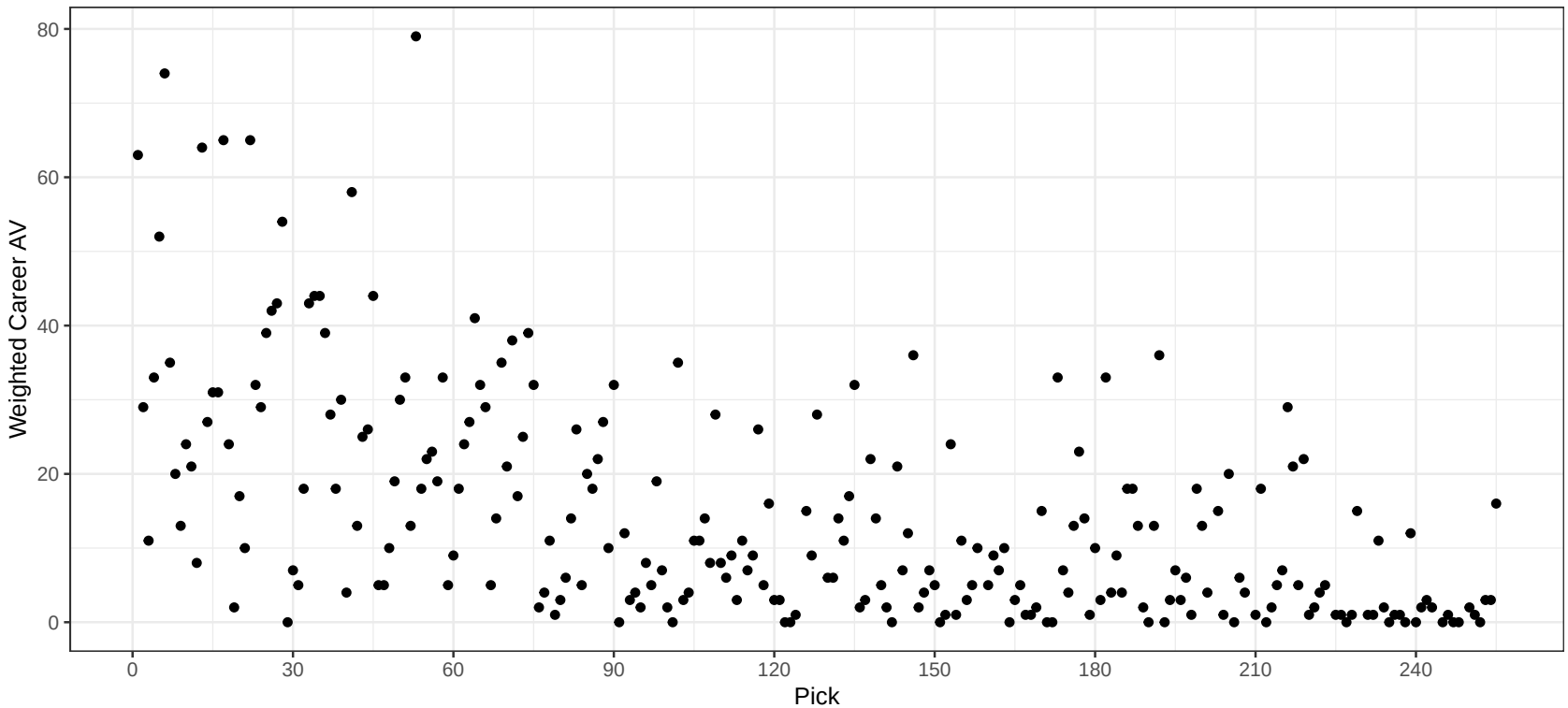
# NFL Player Values

- Approximate value (AV) is an attempt to quantify the value of an NFL player based upon their individual statistics, team performance, and professional awards.
  - A typical starter has AV between 10-20, while an elite player might have AV of 30+



# NFL Player Values

Is a player's career AV related to how high they were picked in the NFL draft?



# Describing Relationships (Qualitatively)

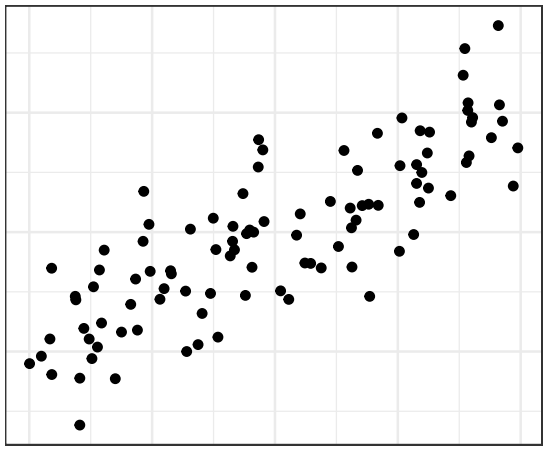
There are *three* important aspects of a relationship between two quantitative variables that we should consider:

1. **Direction** - positive, negative, or unclear
2. **Form** - linear or non-linear
3. **Strength** - weak, moderate, strong, etc.

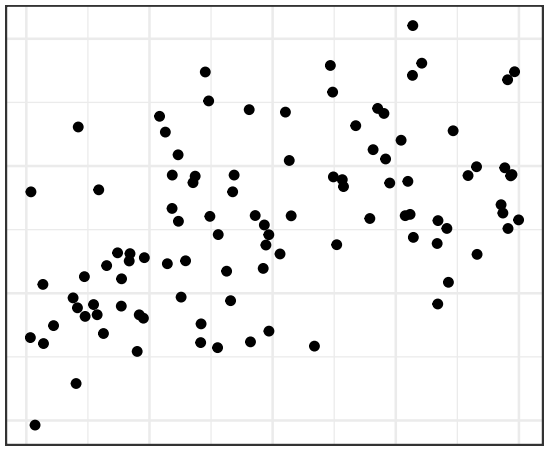
For example: "the resting heart rate and age have a weak, positive, linear association"

# Examples

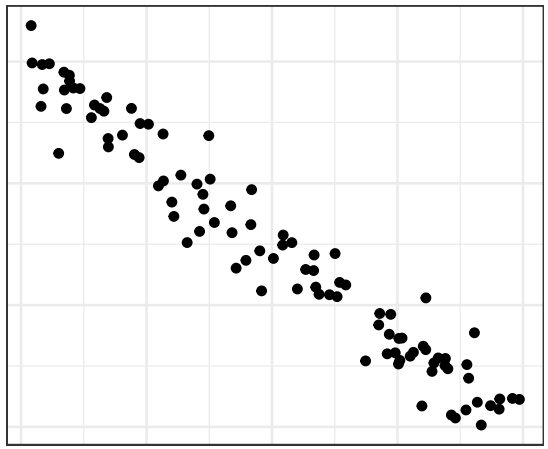
Moderate Positive Linear



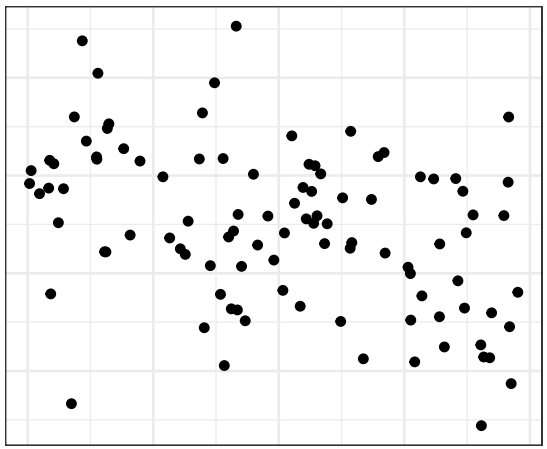
Weak Positive Linear



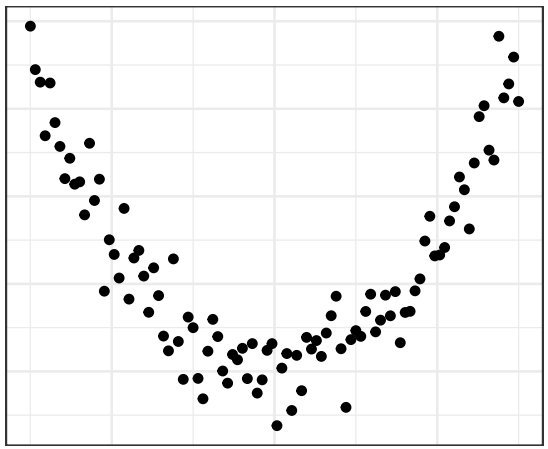
Strong Negative Linear



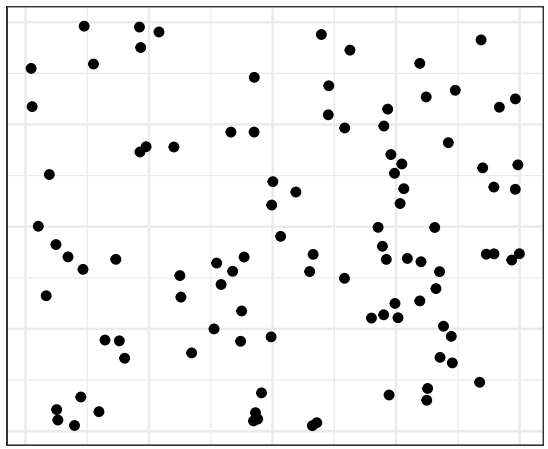
Weak Negative Linear



Strong Nonlinear



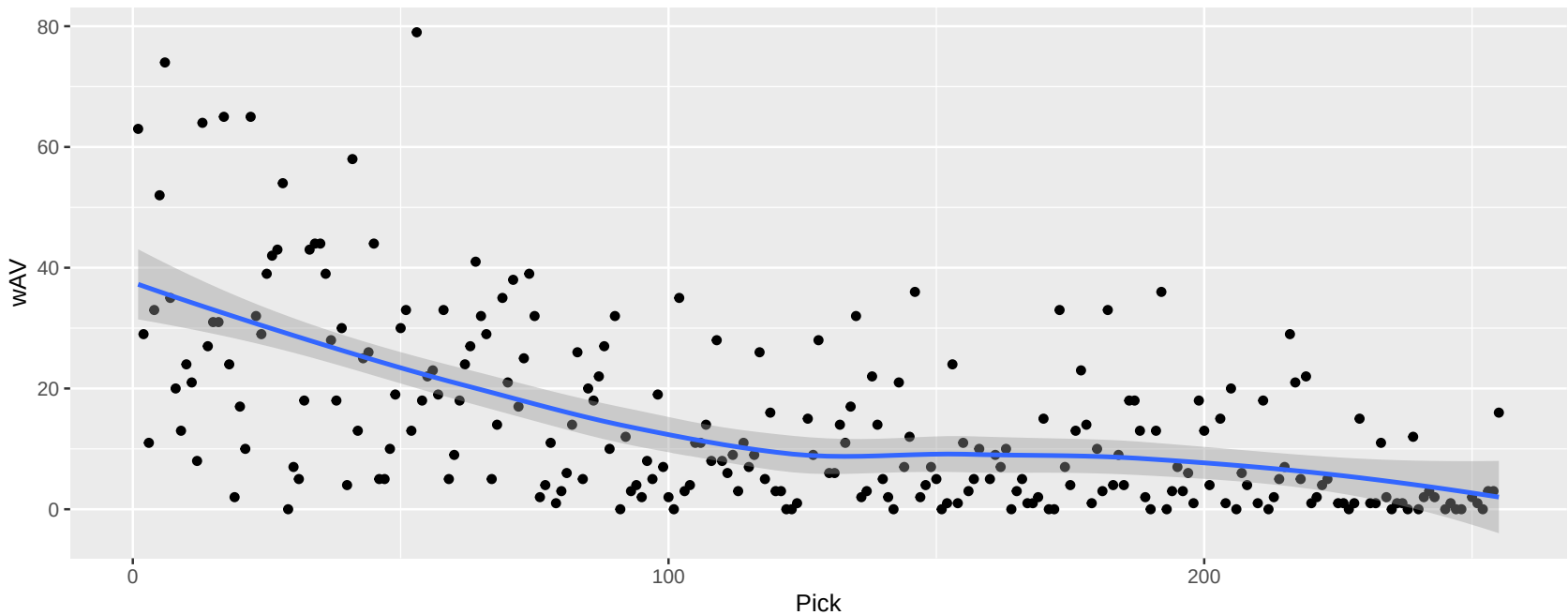
No Clear Relationship



# LOESS

*LOESS* (locally estimated scatterplot smoothing) is a tool to help us judge form:

```
ggplot(data = draft_data, aes(x = Pick, y = wAV)) +
  geom_point() + geom_smooth(method = "loess")
```



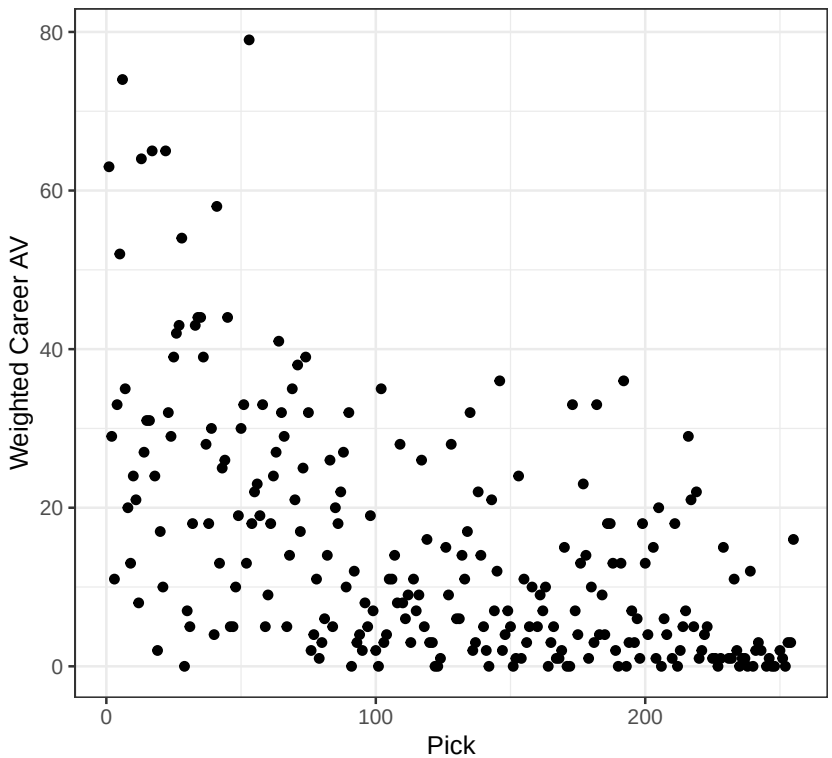
# Practice

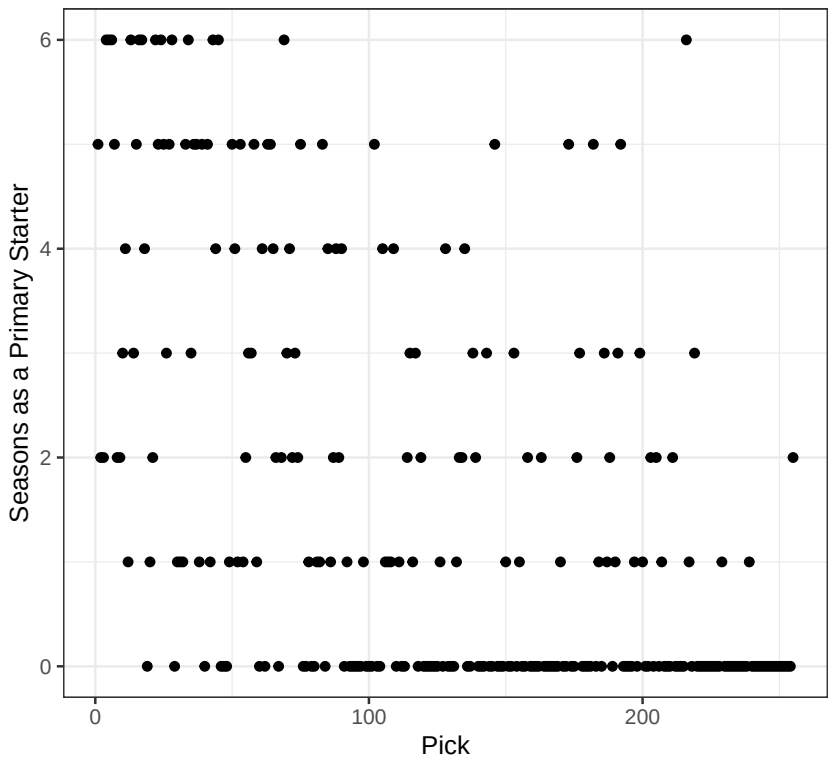
1. Load data into `R` using code below
2. Use the code on the previous slide as a template to make a scatterplot with Loess smoother
3. Describe the direction, form, and strength of the relationship between `Ret_Rate` (percentage of 1st years who return for a 2nd year) and `Med_Grad_Earnings_4Y` (median annual earnings 4 years after graduation)

```
grinnell_peers = read.csv("https://remiller1450.github.io/data/grinnell_peers.csv")
```

# Quantifying Strength

Is an NFL player's draft pick *more strongly* related to their career AV or the number of seasons they play as a starter?



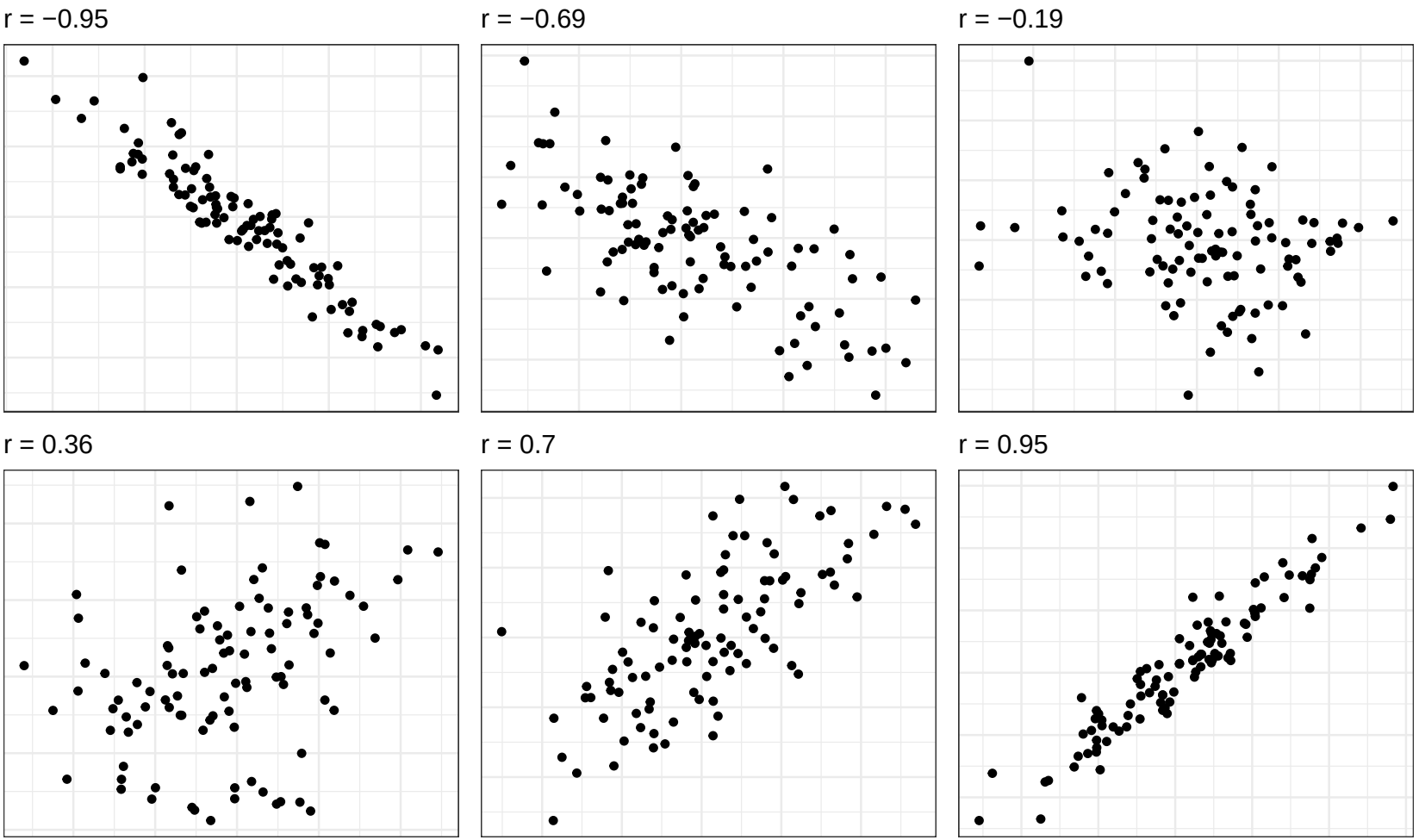
# Quantifying Strength

Karl Pearson developed **Pearson's correlation coefficient** to quantify the strength of *linear association*:

$$r = \frac{1}{n-1} \sum_{i=1}^{n} \left( \frac{x_i - \overline{x}}{s_x} \right) \left( \frac{y_i - \overline{y}}{s_y} \right)$$

- Notice $r$ is essentially the average product of Z-scores across observations, so it has no units
- $r$ can range from -1 (perfect negative association) to $+1$ (perfect positive association) with 0 reflecting no linear association between variables

# Correlation Examples



# Correlation Examples

Whether a correlation is considered "strong", "moderate", or "weak" depends upon the field:

| Correlation Coefficient | | Dancey & Reidy (Psychology) | Quinnipiac University (Politics) | Chan YH (Medicine) |
|---|---|---|---|---|
| +1 | −1 | Perfect | Perfect | Perfect |
| +0.9 | −0.9 | Strong | Very Strong | Very Strong |
| +0.8 | −0.8 | Strong | Very Strong | Very Strong |
| +0.7 | −0.7 | Strong | Very Strong | Moderate |
| +0.6 | −0.6 | Moderate | Strong | Moderate |
| +0.5 | −0.5 | Moderate | Strong | Fair |
| +0.4 | −0.4 | Moderate | Strong | Fair |
| +0.3 | −0.3 | Weak | Moderate | Fair |
| +0.2 | −0.2 | Weak | Weak | Poor |
| +0.1 | −0.1 | Weak | Negligible | Poor |
| 0 | 0 | Zero | None | None |

Source: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6107969/

# Returning to the NFL Player Data

Is an NFL player's draft pick more strongly related to their career AV or the number of seasons they play as a starter?

```
cor(x=draft_data$Pick, y=draft_data$wAV, use = "complete.obs")
```

```
## [1] -0.5617705
```

```
cor(x=draft_data$Pick, y=draft_data$St, use = "complete.obs")
```

```
## [1] -0.5391188
```

# Practice

1. Use the "Grinnell Peers" data set from earlier
2. Is `Ret_Rate` more strongly correlated with `Med_Grad_Earnings_4Y` or with `Enrollment` (total number of enrolled students)?

# Problematic Correlation Examples

From Cook & Swayne's *Interactive and Dynamic Graphics for Data Analysis*:

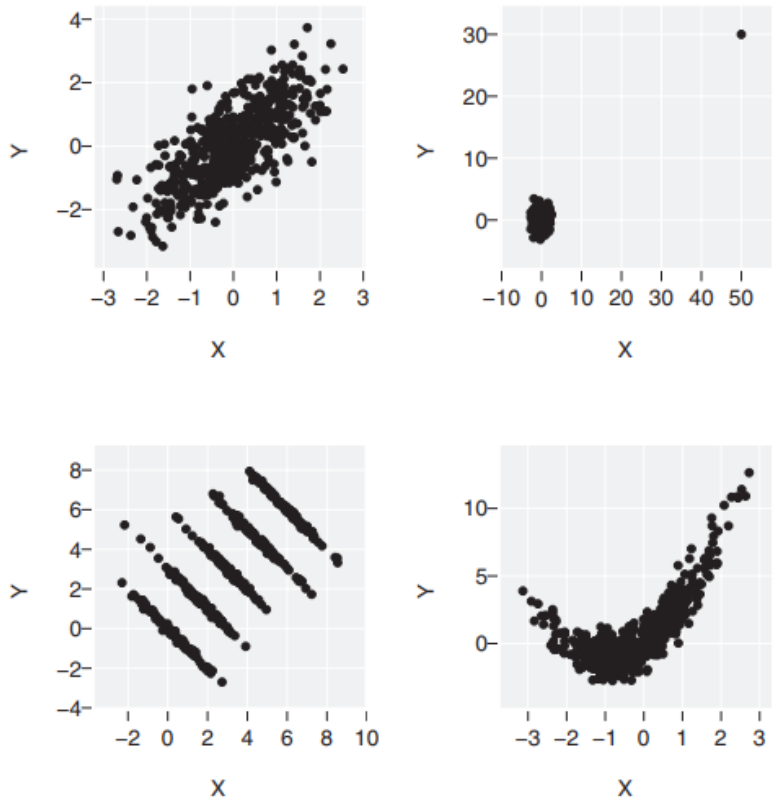



**Fig. 6.1.** Studying dependence between X and Y. All four pairs of variables have correlation approximately equal to 0.7, but they all have very different patterns. Only the top left plot shows two variables matching a dependence modeled by correlation.

# Quantifying Strength for Non-linear Associations

A popular alternative to Pearson' correlation is **Spearman's correlation**, which replaces each observation with its *rank* from smallest to largest and computes the Pearson correlation of those ranks. As an example:

$$X = \{2, 4, 6, 9, 8\}$$
$$Y = \{7, 4, 1, 5, 3\}$$

Replace each value with its rank:

$$X_{\text{rank}} = \{1, 2, 3, 5, 4\}$$
$$Y_{\text{rank}} = \{5, 3, 1, 4, 2\}$$

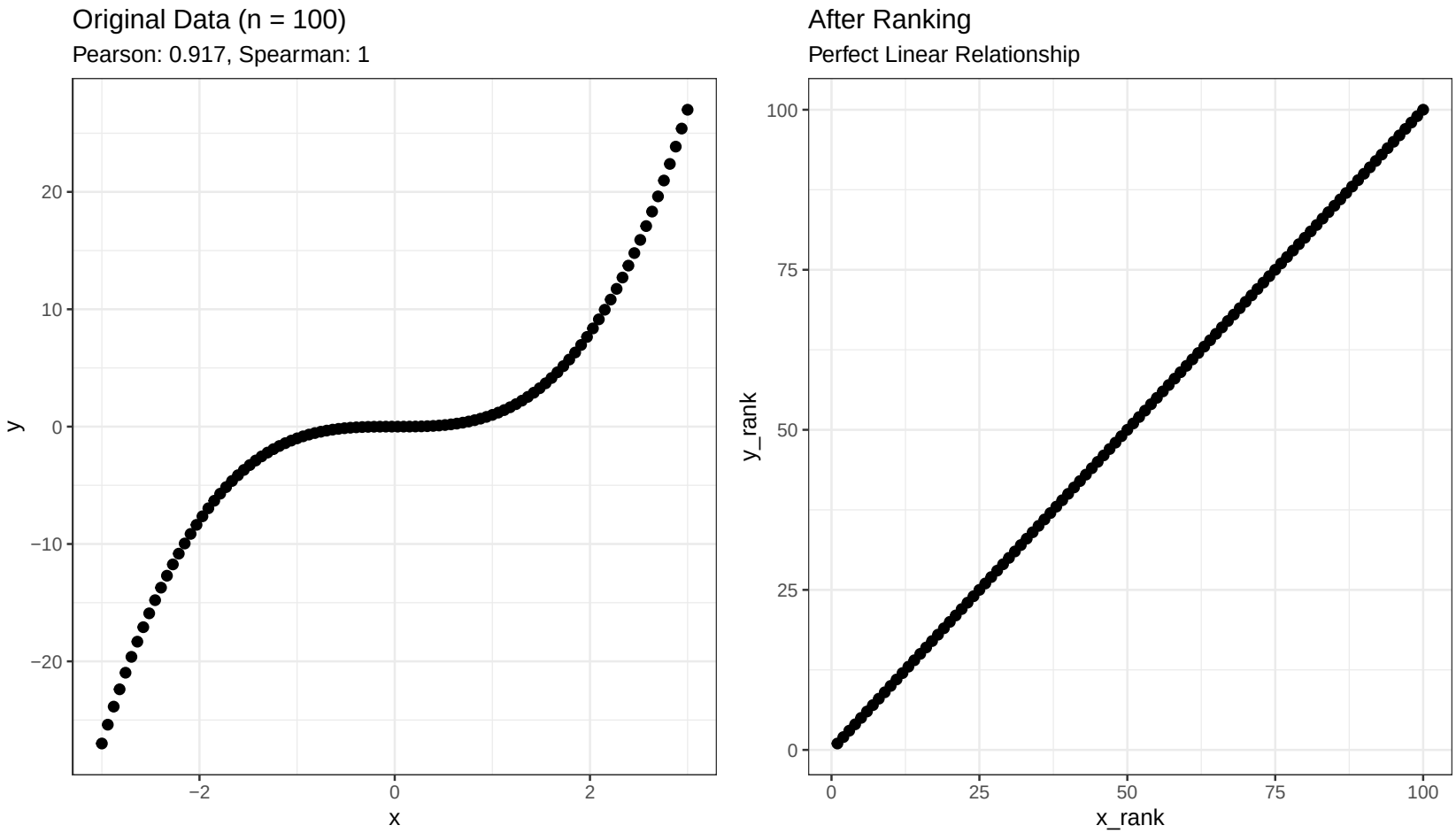
Then correlate the ranks: $\text{cor}(X_{\text{rank}}, Y_{\text{rank}}) = -0.3$

# Quantifying Strength for Non-linear Associations

- **Pearson's correlation** measures the strength and direction of a *linear relationship*.
- **Spearman's correlation** measures the strength and direction of a *monotonic relationship*.

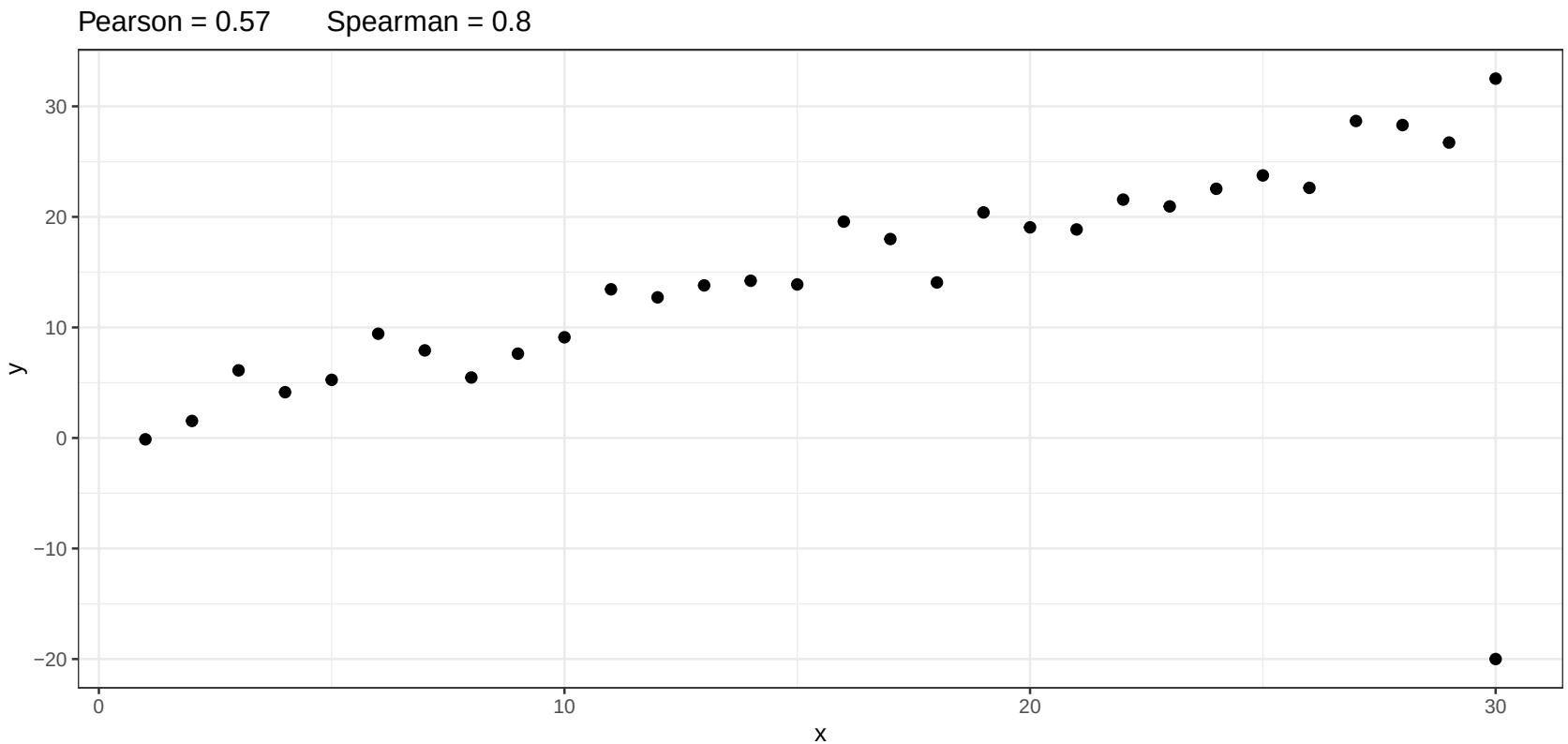
*Note*: A relationship is *monotonic* if larger values of one variable tend to be associated with larger (or smaller) values of the other, even if the relationship is not linear.

# Spearman's Correlation



# Spearman's Correlation

Spearman's correlation is more robust to outliers than Pearson's correlation:



# Practice

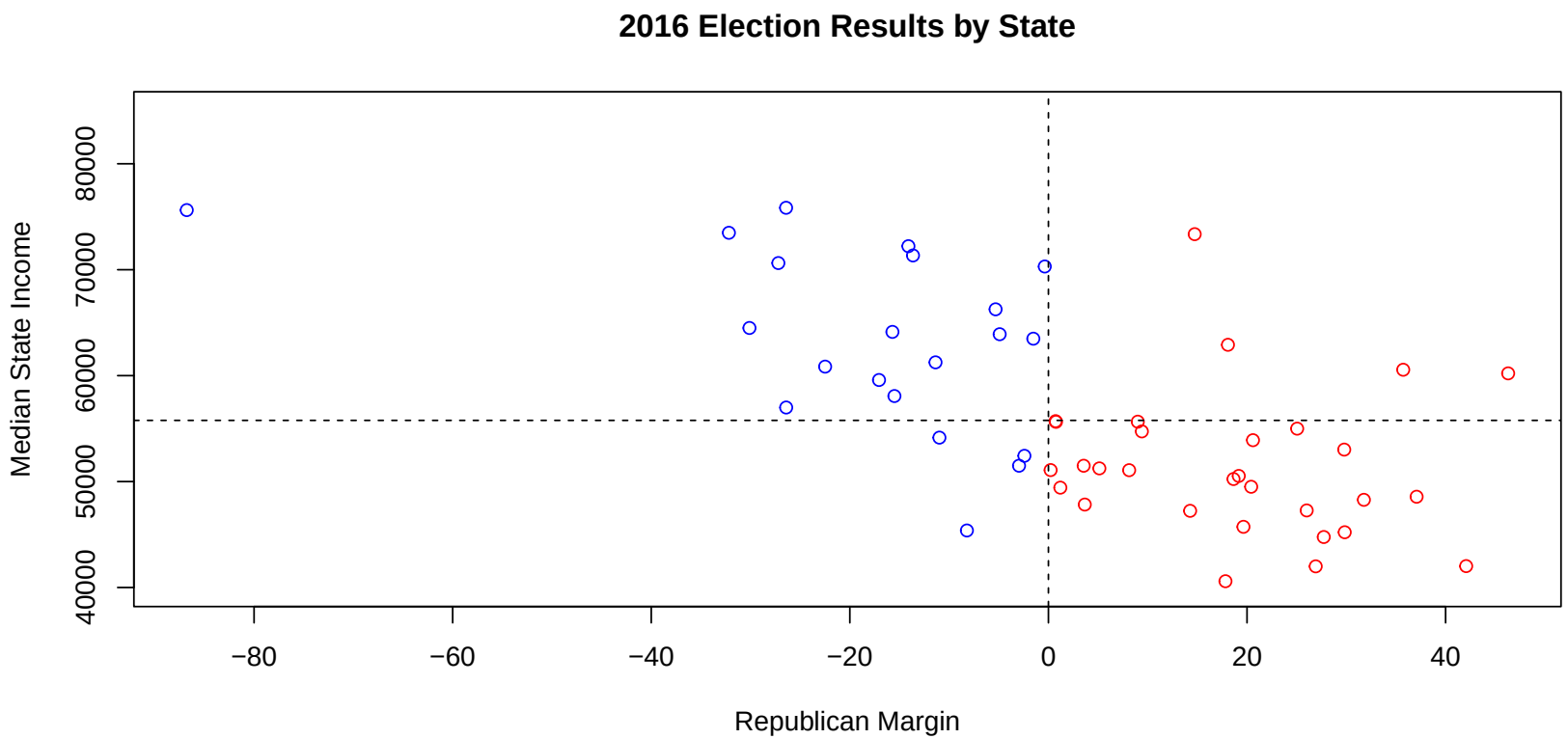
1. Spearman's correlation can be found using the argument `method = "spearman"` in the `cor()` function (see example below). Find Spearman's correlation between the variables `Adm_Rate` and `med_ACT` in the Grinnell peers data set
2. Pearson's correlation between these two variables is -0.824, create a scatterplot of the data and use it to explain why Pearson's and Spearman's correlations show noticeably different degrees of strength.

```
## Example
cor(x=draft_data$Pick, y=draft_data$St, method="spearman")
```

```
## [1] -0.5515649
```

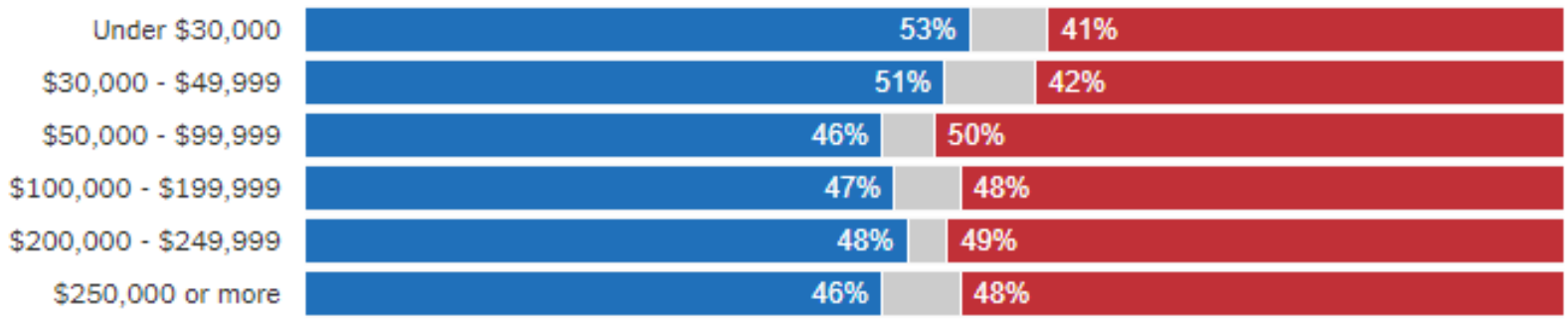
# Other Correlation Misconceptions

Below are state-level data for vote margin in the 2016 presidential election and median household income. Note that $r = -0.63$, so do Republican voters tend to have lower incomes than Democrat voters?



# Ecological Fallacy

Using 2016 exit polls, conducted by the NY Times (Link), we can get a sense of how party vote and income are related *for individuals*:




The **ecological fallacy** occurs when conclusions about *individuals* are made using aggregated (ecological) data. We cannot make reliable conclusions about individuals using aggregate data.

# Conclusion

- We use scatterplots to visualize and qualitatively understand relationships between two quantitative variables
- Correlation is used quantify the strength of relationship
  - Pearson's correlation for linear relationships
  - Spearman's correlation for non-linear (but monotonic) relationships
- The ecological fallacy frequently arises in correlation analysis, so you should be careful when analyzing aggregate data (ie: state-level, country-level, etc.)
- Another common mistake in correlation analysis is inferring that a strong correlation implies a causal relationship (we'll talk more about causation soon)